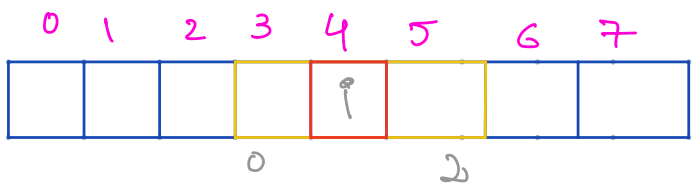
# Background

$$y_i = \sum_{j=0}^{2r} f_j * x_{i+j-r}$$




$i - j$

$2(1)+1 = 3$

$4+0-1 = 3$

$4+1-1 = 4$

$4+2-1 = 5$

$$P_{y,x} = \sum_{i=0}^{2r_y} \sum_{j=0}^{2r_x} f_{i,j} * N_{y-r_y+i,\,x-r_x+j}$$

# Parallel convolution: a basic algorithm

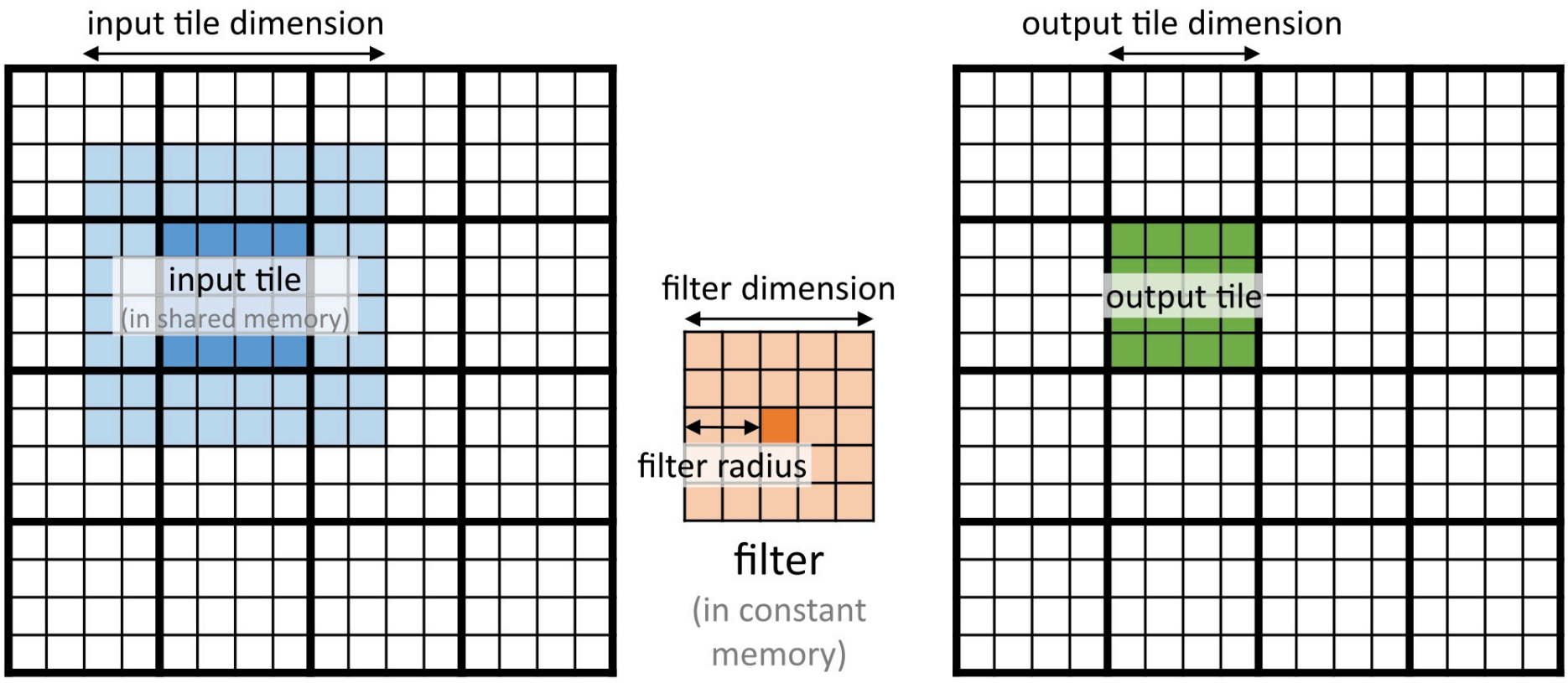
```
01 __global__ void convolution_2D_basic_kernel(float *N, float *F, float *P,
       int r, int width, int height) {
02     int outCol = blockIdx.x*blockDim.x + threadIdx.x;
03     int outRow = blockIdx.y*blockDim.y + threadIdx.y;
04     float Pvalue = 0.0f;
05     for (int fRow = 0; fRow < 2*r+1; fRow++) {
06       for (int fCol = 0; fCol < 2*r+1; fCol++) {
07           inRow = outRow - r + fRow;
08           inCol = outCol - r + fCol;
09           if (inRow >= 0 && inRow < height && inCol >= 0 && inCol < width) {
10              Pvalue += F[fRow*(2*r+1)+fCol]*N[inRow*width + inCol];
11           }
12       }
13     }
14     P[outRow][outCol] = Pvalue;
15 }
```

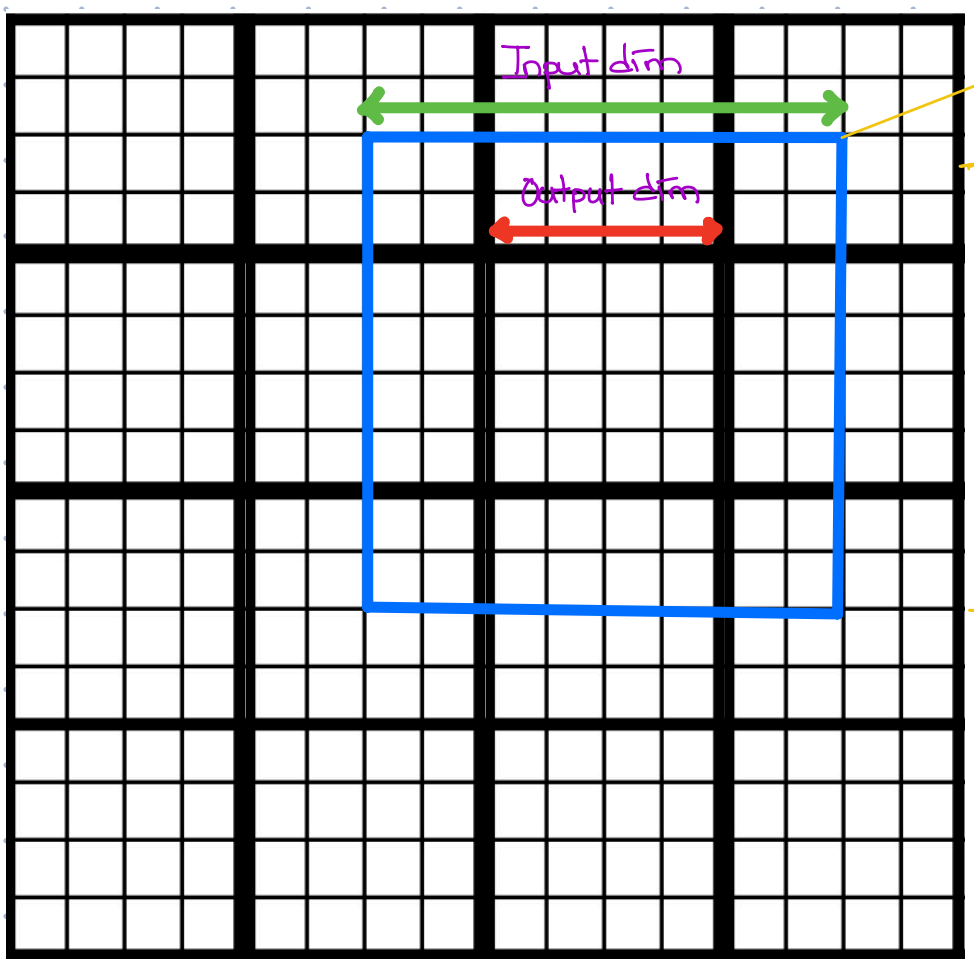
FLOP — 4 bytes — FLOP — 4 bytes

$$Flops/B = \frac{4+4}{1+1}$$

$$= 0.25$$

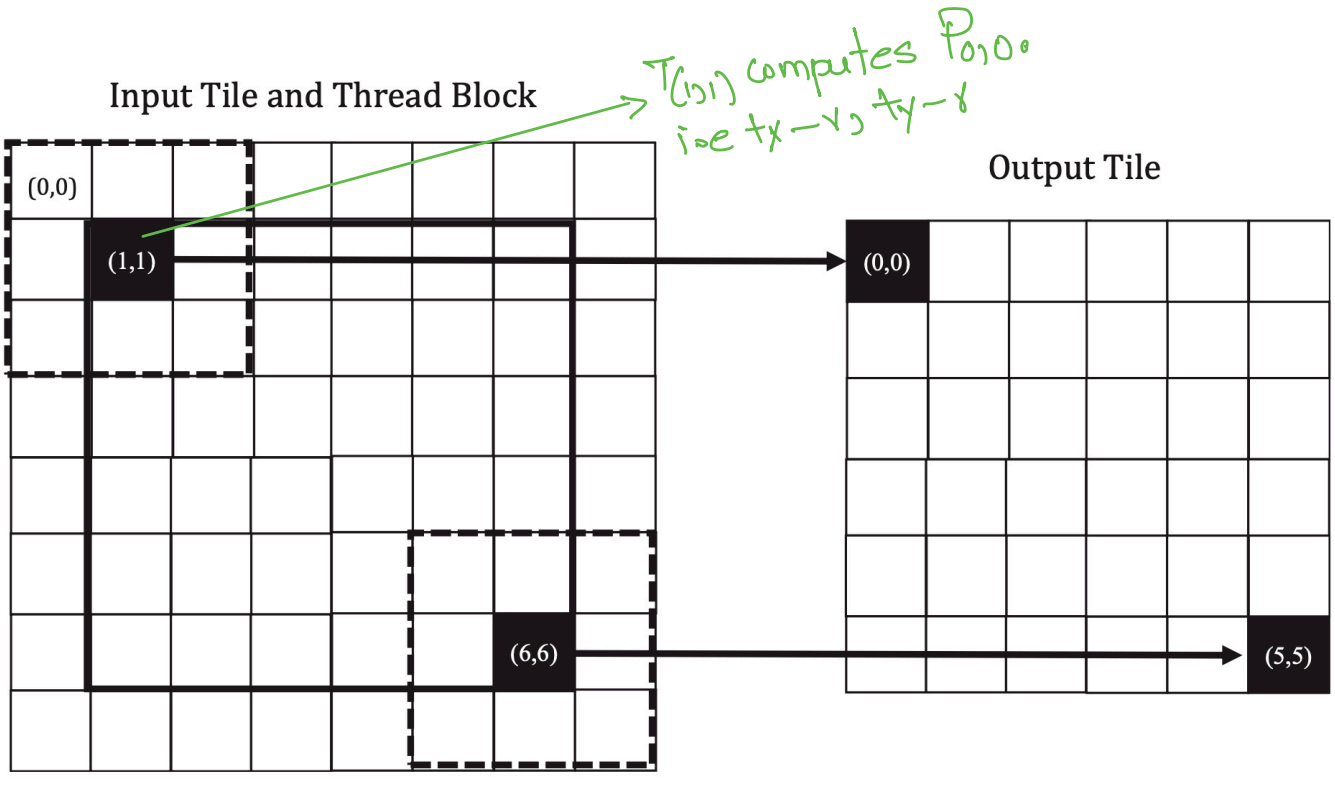
# Tiled convolution with halo cells

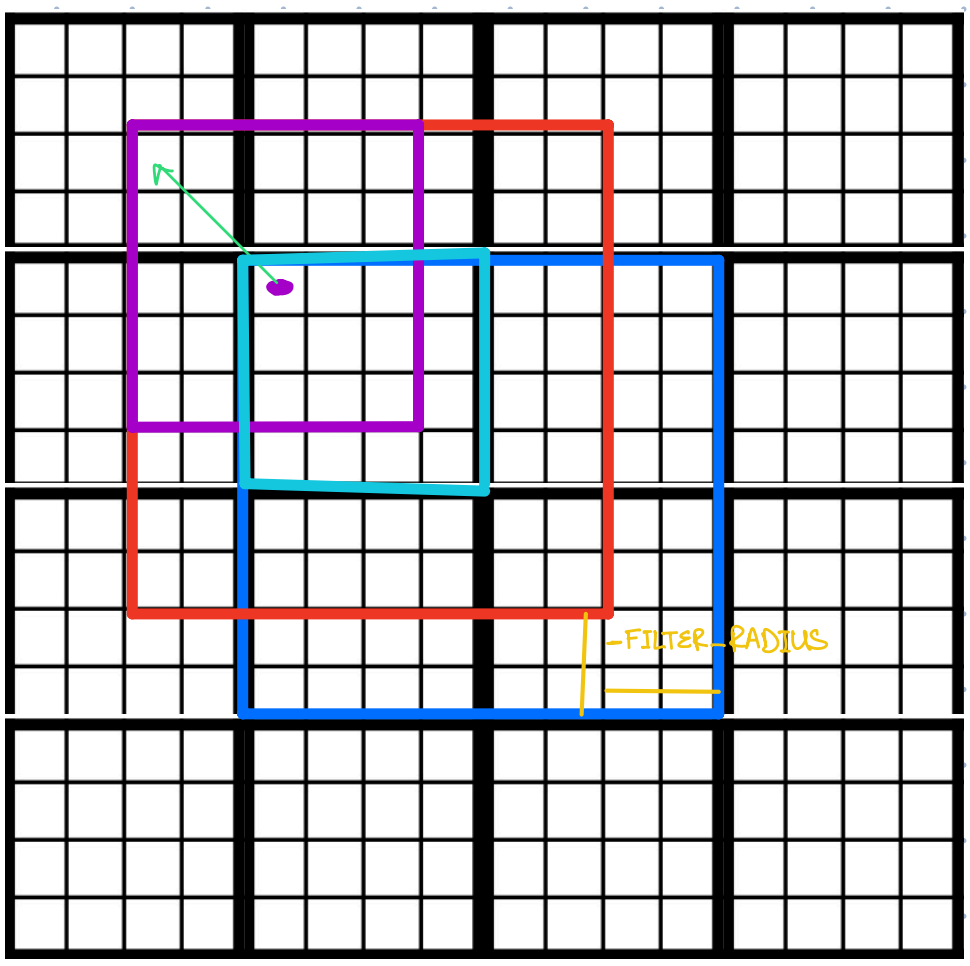



A thread block is launched.

Within this block, only threads in this block is active. Other are disabled i.e threads whose index is in filter radius.




FILTER_RADIUS